MIL-E-1/1421A
27 April 1972
SUPERSEDING
MIL-E-1/1421 (Sig C)
12 October 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5844

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, miniature

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	2a	1a	h	h	1g	2g	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	ec	Ehk	Rg/g	Rk	Ib/a	TE	Alt
Unit:	V	Vdc	Vdc	v	v	Meg	Ohms	mAdc	°C	ft
Maximum:	6.6	200	0	---	100	0.05 (Note 1)	---	10	120	Note 2
Minimum:	6.0	---	-75	-150	---	---	---	---	---	---
TEST CONDITIONS:	6.3	100	0	---	---	---	470	---	---	---

GENERAL:

(A) Qualification - Not required

(A) denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	275	325	mA
1266	Total grid current	4, 5	Rg = 0.5 Meg	0.65	II	Ic	---	-1.0	μAdc
1256	Electrode current (1) (anode)	5	Ebb = 150 Vdc; Rk = 0; Rg/g = 0.047 Meg; Rp/a = 0.02 Meg	0.65	II	Ib	5.0	---	mAdc
1256	Electrode current (2) (anode)	4, 5	Ebb = 150 Vdc; Ec = -12 Vdc; Rk = 0; Rg/g = 0.047 Meg; Rp/a = 0.02 Meg	0.65	II	Ib	---	100	μAdc
1201	Short and discontinuity detection	5		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	5		2.5	I	R	100	---	Meg
1336	Heater-cathode leakage	-		2.5	I	Ihk	---	20	μAdc
1256	Electrode current (3) (anode)	5	Ef = 5.7; Rk = 0; Ebb = 150 Vdc; Rg/g = 0.047 Meg; Rp/a = 0.02 Meg	---	---	Ib	4.6	---	mAdc
---	Bias balance	6	Ebb = 150 Vdc; Rk = 0; Rg/g = 0.047 Meg; Rp/a = 0.020 Meg	2.5	I	Ec	---	2.0	Vdc
1256	Electrode current (4) (anode)	5		---	---	Ib	3.8	5.8	mAdc
1306	Transconductance	5		---	---	Gm	2,700	4,700	μmhos
1231	Emission	5	Eb = Ec = 10 Vdc; Rk = 0	2.5	I	Ie	40	---	mAdc
1331	Direct-interelectrode capacitance	-	No shield	6.5	Code E	Cgp	2.1	3.1	pF
(A)						Cin	2.1	3.1	pF
						Cout (sec 1)	0.25	0.75	pF
						Cout (sec 2)	0.13	0.63	pF
						Cgp	---	1.00	pF
1031	Low-frequency (25 Hz) vibration	7	Ebb = 250 Vdc; Ec = -8 Vdc; Rk = 0; Rp = 2,000 ohms	6.5	Code H	Ep	---	200	mVac
1041	Shock	7, 8	450 G; Eb = 300 Vdc; Ec = -8 Vdc; Ehk = +100 Vdc; Rg = 0.1 Meg; Rk = 0	---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 - Continued								
---	Post-shock test end points:								
1201	Short and discontinuity detection	-		---	---	---	---	---	---
1031	Low frequency (25 Hz) vibration	-		---	---	Ep	---	300	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1256	Electrode current (1) (anode)	-		---	---	Ib	4.7	---	mAdc
1266	Total grid current	-		---	---	Ic	---	-1.5	µAdc
(A)1121	Base strain	-		---	---	---	---	---	---
(A)2126	Glass strain	-		2.5	I	---	---	---	---
(A)1106	Permanence of marking	-		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3										
1506	Heater-cycling life	-	Ef = 7.5 V; Ehk = +135 Vdc; Ec = Eb = 0	---	---	---	---	---	---	---	---
---	Heater-cycling life-test end point:										
1336	Heater-cathode leakage	-		---	---	---	---	Ihk	---	60	µAdc
---	Life test (zero bias)	9	Eb = 150 Vdc; Ehk = +100 Vdc; Rg/g = 0.047 Meg; Rp/a = 0.02 Meg; Rk = 0	---	---	---	---	---	---	---	---
---	Life-test (zero bias) end points (1,000 hours):										
---	Inoperatives	-		---	---	2	4	---	---	---	---
---	Cathode interface resistance	10	Ef = 5.7 Vdc; Eb = 75 Vdc; Ec/Ib = 2.0mAdc; Rk = 0	---	---	2	4	Ri	---	25	Ohms
1266	Total grid current	-		---	---	2	4	Ic	---	-2.0	µAdc
1256	Electrode current (1) (anode)	-		---	---	2	4	Ib	4.7	---	mAdc
1336	Heater-cathode leakage	-	Ehk = -100 Vdc	---	---	3	6	Ihk	---	30	µAdc
1811	Insulation of electrodes	-		---	---	3	6	R	10	---	Meg
---	Total defectives	3		---	---	5	8	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	MSF LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
						1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 5 - Continued										
---	Life test (cut off)	9	Eb = 150 Vdc; Ecc = -75 Vdc; Ehk = -100 Vdc; Rg/g = 0.07 Meg; Rp/a = 0.02 Meg; Rk = 0.02 Meg	---	---	---	---	---	---	---	---
---	Life-test (cut off) end points (1,000 hours):										
---	Inoperatives	-		---	---	2	4	---	---	---	---
---	Cathode interface resistance	10	Ef = 5.7 Vdc; Eb = 75 Vdc; Ec/Ib = 2.0 mAdc	---	---	2	4	Ri	---	25	Ohms
1266	Total grid current	-		---	---	2	4	Ic	---	-2.0	µAdc
1256	Electrode current (1) (anode)	-		---	---	2	4	Ib	4.7	---	mAdc
1336	Heater-cathode leakage	-	Ehk = -100 Vdc	---	---	3	6	Ihk	---	30	µAdc
1211	Insulation of electrodes	-		---	---	3	6	R	10	---	Meg
---	Total defectives	3		---	---	5	8	---	---	---	---

NOTES:

1. This value is for operation under fixed-bias conditions; with cathode bias, Rg/g may be 0.1 Meg (max).
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. The total defectives for the zero-bias life and cut-off life operation tests, combined, shall not exceed the total defectives value specified. If the value specified for the allowable defects per characteristic is exceeded, the sample under the particular test shall be considered to have failed that test.
4. This test to be performed at the conclusion of the holding period.
5. Test each unit separately with opposite unit grounded.
6. Vary E1c and E2c for Ib1 = Ib2 = 100 µAdc; the absolute value of the difference between E1c and E2c shall not exceed the maximum limit specified.
7. With 1g tied to 2g, and 1a tied to 2a.
8. A grid resistor of 0.1 Meg shall be added; however, this resistor shall not be used when a thyratron-type indicator is used.
9. The test sample shall consist of 20 tubes, selected at random, for the zero-bias life-test operation, and 20 tubes also selected at random for the cut-off life-test operation. In the event of failure of the first sample, a second sample of 20 tubes each for the zero-bias and cut-off life-test operations shall be selected at random, and subjected to these 2 life tests. Acceptance shall then be based on the combined results from the first and second samples (80 tubes).
10. Interface resistance shall be measured immediately after short and discontinuity detection test and before any other life end-point tests. Cathode interface resistance shall be measured in accordance with test method 1511.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES
Other - NS

User activities:
Army - MI, WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2620)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: AMSEL-PP-ED
Fort Monmouth, New Jersey 07703

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/1021A ELECTRON TUBE, RECEIVING TYPE 5814

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FN 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:40Z
Check the source to verify that this is the current version before use.